

## ORIGINAL ARTICLE

## Inference of interactions in cyanobacterial–heterotrophic co-cultures via transcriptome sequencing

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**We used deep sequencing technology to identify transcriptional adaptation of the euryhaline unicellular cyanobacterium *Synechococcus* sp. PCC 7002 and the marine facultative aerobic *Shewanella putrefaciens* W3-18-1 to growth in a co-culture and infer the effect of carbon flux distributions on photoautotroph–heterotroph interactions. The overall transcriptome response of both organisms to co-cultivation was shaped by their respective physiologies and growth constraints. Carbon limitation resulted in the expansion of metabolic capacities, which was manifested through the transcriptional upregulation of transport and catabolic pathways. Although growth coupling occurred via lactate oxidation or secretion of photosynthetically fixed carbon, there was evidence of specific metabolic interactions between the two organisms. These hypothesized interactions were inferred from the excretion of specific amino acids (for example, alanine and methionine) by the cyanobacterium, which correlated with the downregulation of the corresponding biosynthetic machinery in *Shewanella* W3-18-1. In addition, the broad and consistent decrease of mRNA levels for many Fe-regulated *Synechococcus* 7002 genes during co-cultivation may indicate increased Fe availability as well as more facile and energy-efficient mechanisms for Fe acquisition by the cyanobacterium. Furthermore, evidence pointed at potentially novel interactions between oxygenic photoautotrophs and heterotrophs related to the oxidative stress response as transcriptional patterns suggested that *Synechococcus* 7002 rather than *Shewanella* W3-18-1 provided scavenging functions for reactive oxygen species under co-culture conditions. This study provides an initial insight into the complexity of photoautotrophic–heterotrophic interactions and brings new perspectives of their role in the robustness and stability of the association.**

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## Introduction

Experimental evidence suggests the existence of strong positive and negative interactions between photoautotrophic and heterotrophic microorganisms (Caldwell and Caldwell, 1978; Cole, 1982; Carpenter and Foster, 2002; Amin *et al.*, 2012).

In aquatic environments, an important class of interactions is based on cross-feeding and metabolite exchange, whereby photosynthetically fixed dissolved organic carbon can elicit chemotactic responses (Paerl and Gallucci, 1985; Seymour *et al.*, 2010) that lead to the development of spatial associations (Paerl and Pinckney, 1996; Bertilsson *et al.*, 2007). The spectrum of behavioral responses varies greatly, as the origin of excreted material ranges from targeted secretion of photosynthetic intermediates (for example, glycolate, osmolytes and fatty acids) and extracellular polymeric substance (Seymour *et al.*, 2010; Bruckner *et al.*, 2011), to the products of cell lysis that can include sugars, proteins, lipids and nucleic acids (Grossart, 1999;

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Stevenson and Waterbury, 2006; Shen *et al.*, 2011). In exchange, heterotrophic bacteria are thought to provide essential micronutrients, such as vitamins, amino acids and bioavailable trace metals (Amin *et al.*, 2009; Hayashi *et al.*, 2011; Kazamia *et al.*, 2012; Xie *et al.*, 2013), necessary to maintain high photosynthetic productivity.

Furthermore, tight associations, which involve complementation of physiological or biochemical functions, may undergo evolutionary selection emphasizing optimal metabolic performance and enhanced productivity (Overmann, 2006; Stevenson and Waterbury, 2006). For example, stimulation of cyanobacterial N<sub>2</sub> fixation in association of *Anabaena* sp. with heterotrophic bacteria is well known and attributed to the respiratory O<sub>2</sub> removal during periods of high ambient oxygen concentrations (Paerl and Kellar, 1978). Similarly, co-cultivation of marine heterotrophs with *Prochlorococcus* ecotypes has been shown to enhance photosynthetic growth (Morris *et al.*, 2008; Sher *et al.*, 2011). This positive effect was attributed to the decrease of oxidative stress by heterotrophs through catalase-mediated scavenging of peroxide and other reactive oxygen species (ROS) (Morris *et al.*, 2008). Such interactions have ecosystem-scale consequences because ROS may directly affect nutrient bioavailability, photosynthetic productivity and carbon fluxes (Amin *et al.*, 2009).

As we move toward a system-level understanding of community organization and function (Zengler and Palsson, 2012), the behavioral complexity increases with the number of interacting organisms (Wintermute and Silver, 2010). To that end, application of genome-scale approaches to dissect subcellular pathways and regulatory networks involved in governing interspecies interactions is dependent, at least initially, on the availability of model biological systems and genomic information (Tai *et al.*, 2009). Here, we investigated the growth of the euryhaline unicellular cyanobacterium *Synechococcus* sp. PCC 7002 (hereafter *Synechococcus* 7002) in a co-culture with a marine facultative aerobe *Shewanella putrefaciens* W3-18-1 (hereafter *Shewanella* W3-18-1) under different trophic conditions. Although recognizing the opportunistic nature of interactions between these two strains, there is ample evidence that certain *Shewanella* species live in association with photoautotrophic prokaryotes (Simidu *et al.*, 1990; Bowman *et al.*, 1997; Salomon *et al.*, 2003). Through the application of next-generation sequencing technology in conjunction with controlled cultivation and metabolomic profiling, we were able to delineate specific transcriptional responses of each organism to co-cultivation. Building upon previous genome-scale investigations of phototrophic consortia (Tai *et al.*, 2009; Wenter *et al.*, 2010), the functional annotation of the differentially regulated transcripts leads to the development of a conceptual model of interactions between

cyanobacteria and heterotrophs as a function of carbon sources and flux directions.

## Materials and methods

### Strains and growth conditions

Axenic batch cultures of *Synechococcus* 7002 were routinely grown in modified basal medium A containing: 18 g l<sup>-1</sup> NaCl, 0.6 g l<sup>-1</sup> KCl, 0.9 g l<sup>-1</sup> NH<sub>4</sub>Cl, 5.0 g l<sup>-1</sup> MgSO<sub>4</sub> · 7 H<sub>2</sub>O, 50 mg l<sup>-1</sup> KH<sub>2</sub>PO<sub>4</sub>, 266 mg l<sup>-1</sup> CaCl<sub>2</sub>, 30 mg l<sup>-1</sup> Na<sub>2</sub>EDTA · 2 H<sub>2</sub>O, 3.89 mg l<sup>-1</sup> FeCl<sub>3</sub> · 6 H<sub>2</sub>O, 1 g l<sup>-1</sup> Tris HCl (pH 8.2), 34.26 mg l<sup>-1</sup> H<sub>3</sub>BO<sub>3</sub>, 4.32 mg l<sup>-1</sup> MnCl<sub>2</sub> · 4 H<sub>2</sub>O, 0.315 mg l<sup>-1</sup> ZnCl<sub>2</sub>, 0.03 mg l<sup>-1</sup> MoO<sub>3</sub>, 12.15 µg l<sup>-1</sup> CoCl<sub>2</sub> · 6 H<sub>2</sub>O, 3 µg l<sup>-1</sup> CuSO<sub>4</sub> · 5 H<sub>2</sub>O and 4 µg l<sup>-1</sup> vitamin B<sub>12</sub> (Stevens and Porter, 1980; Ludwig and Bryant, 2011). The cultivation was carried out at room temperature in 1-L Roux bottles sparged with 2% CO<sub>2</sub>-enriched air under continuous 50 µmol m<sup>-2</sup> s<sup>-1</sup> illumination. Axenic *Shewanella* W3-18-1 cultures were maintained at 30 °C in tryptic soy broth or M1 defined medium (Pinchuk *et al.*, 2008) amended with 20 mM D,L-Na lactate and 10 ml each of 10X Wolfe's vitamin and 10X mineral solutions (Kieft *et al.*, 1999).

Continuous cultivation experiments were carried out in a custom-made photobioreactor built using the Bioflo 3000/310 fermenter platform (New Brunswick Scientific, New Brunswick, NJ, USA). The system consisted of a 7.5-l borosilicate glass vessel (49.5 cm height, 13.4 cm inner diameter) and a cage-like light enclosure carrying eight 14-W Pentron 3500K tubular fluorescent lights (model FP14/835/ECO, Osram Sylvaania, Danvers, MA, USA) mounted in pairs at each quadrant of the vessel. Irradiance was controlled using a custom dimmer integrated within the BioFlo 310 bioreactor controller module. A solar blanket was used to isolate the bioreactor from ambient light. Control over other cultivation parameters including temperature, agitation, pH and dissolved O<sub>2</sub> tensions was achieved using built-in BioFlo 310 controls. Temperature was set at 30 °C, agitation at 150 r.p.m. and pH of 7.5 was maintained throughout the run by automatic bases or acid addition. Dissolved O<sub>2</sub> tension, expressed as percent of O<sub>2</sub> saturation, was continuously monitored with an InPro 6110/320 DO probe (Mettler-Toledo Inc., Toledo, OH, USA) and maintained at a desired value by automatically altering the proportion of O<sub>2</sub> and air.

### Chemostat cultivation

All growth studies, expression profiling and metabolomics measurements were conducted in axenic and mixed cultures grown under steady-state conditions. Chemostat cultivation was specifically chosen to provide a well-defined physiological state, which eliminates confounding variables that often complicate transcriptome analyses (Wu *et al.*, 2004; Bull, 2010). In chemostats, *Synechococcus* 7002 and *Shewanella* W3-18-1 were grown in the

basal medium A in the absence of Tris HCl using 8 mM  $\text{NaHCO}_3$  and 5 mM D,L-Na lactate, respectively, as the sole source of carbon. Co-culture chemostats were initially established by adding 0.5 l of lactate-grown *Shewanella* W3-18-1 cells to a 5.5-l chemostat culture of *Synechococcus* 7002 and a gradual transition from  $\text{NaHCO}_3$  to D,L-Na lactate as the sole source of carbon. Transition of the co-culture back to growth with  $\text{NaHCO}_3$  was carried out in a step-wise manner by gradually reducing lactate concentration in the feed to minimize cell lysis and wash-out. The steady state was inferred from stability of the following growth readouts:  $\text{OD}_{730}$  ( $\leq 3\%$  variation between measurements), pH and dissolved  $\text{O}_2$  concentration. Samples for all the analyses were only taken after at least five volume changes under steady-state conditions. The details of media delivery rate, biomass concentration and dissolved  $\text{O}_2$  tensions for each steady-state condition are specified in the text and in Table 1. The irradiance values for each steady state are reported as quanta incident to the center of the reactor. The measurements were carried out using LI193S underwater spherical quantum sensor (LI-COR, Lincoln, NE, USA) and determined to be radially isotropic. The steady-state chemostat conditions used for transcriptome analyses were determined to be light saturated and only carbon limited. Light saturation was assessed by the fact that biomass ( $\text{OD}_{730}$ ) did not respond to small increases or decreases in incident irradiance.

#### Analytical procedures and metabolite detection

Nuclear magnetic resonance spectroscopy (NMR) analysis was used to quantitatively measure excreted organic metabolites in axenic and mixed chemostat cultures of *Synechococcus* 7002. Twenty-milliliter aliquots from each culture were passed through 0.2  $\mu\text{m}$  membranes (EMD Millipore Corporation, Billerica, MA, USA) and spiked with 1/10 volume of a solution containing 5 mM 2,2-dimethyl-2-silapentane-5-sulfonate and 0.2% sodium

azide (Sigma-Aldrich, St Louis, MO, USA) dissolved in 100% deuterium oxide ( $\text{D}_2\text{O}$ , Cambridge Isotope Laboratories, Tewksbury, MA, USA). All samples were stored at 6 °C until data collection. NMR data were acquired on a Varian Direct Drive (VNMR) 600 MHz spectrometer (Agilent Technologies, Santa Clara, CA, USA) equipped with a Dell Precision T3500 Linux workstation (Dell, Plano, TX, USA) running VNMRJ 3.2 (Agilent Technologies). The spectrometer system was outfitted with a Varian triple resonance salt-tolerant cold probe with a cold carbon preamplifier. A Varian standard one-dimensional proton nuclear overhauser effect spectroscopy (NOESY) with presaturation (TNNOSY) was collected on each sample, using the following data collection protocol: nonselective 90° excitation pulse, a 100 ms mixing time, acquisition time of 4 s, a presaturation delay of 1.5 s, spectral width of 12 p.p.m. and temperature control set to 25 °C. Collected spectra were analyzed using Chenomx 7.6 software (Edmonton, Alberta, Canada), with quantifications based on spectral intensities relative to 0.5 mM 2,2-dimethyl-2-silapentane-5-sulfonate.

#### Growth stoichiometry and theoretical carbon balance

Maximum theoretical yields were calculated from stoichiometric equations formulated on a Cmol basis (see Supplementary Material Equations). Equations were derived for the oxidation of lactate and subsequent reduction of inorganic carbon (represented as  $\text{CO}_2$ ) and accounted for both biomass and extracellular metabolite production. Net theoretical yields were calculated from stoichiometric carbon balances.

#### Flow cytometry and imaging

The flow cytometry data were obtained using a BD Influx Fluorescence-Activated Cell Sorter (BD Biosciences, San Jose, CA, USA). Upon harvesting, the cells were immediately treated with 50 mM  $\text{Na}_2\text{EDTA}$  (Sigma-Aldrich) and gently pipetted to

**Table 1** Baseline cultivation conditions and physiological outputs of *Synechococcus* 7002 and *Shewanella* W3-18-1 grown axenically and in co-cultures

| Organism(s) and chemostat condition <sup>a</sup>                              | Carbon source           | Dilution rate, $\text{h}^{-1}$ | Irradiance, $\mu\text{mol photons m}^{-2} \text{s}^{-1 \text{b}}$ | DOT, % of $\text{O}_2$ saturation | $\text{OD}_{730}$ | Biomass, AFDW, $\text{mg l}^{-1}$ |
|-------------------------------------------------------------------------------|-------------------------|--------------------------------|-------------------------------------------------------------------|-----------------------------------|-------------------|-----------------------------------|
| <i>Shewanella</i> W3-18-1 carbon-limited chemostat                            | D,L-lactate, 5 mM       | 0.075                          | 1720                                                              | 50.1                              | 0.105             | 131                               |
| <i>Synechococcus</i> 7002 carbon-limited chemostat                            | $\text{HCO}_3^-$ , 8 mM | 0.075                          | 1720                                                              | 50.2                              | 0.327             | 150                               |
| <i>Synechococcus</i> 7002, <i>Shewanella</i> W3-18-1 carbon-limited chemostat | D,L-lactate, 5 mM       | 0.075                          | 1720                                                              | 49.5                              | 0.431             | 172                               |
| <i>Synechococcus</i> 7002, <i>Shewanella</i> W3-18-1 carbon-limited chemostat | $\text{HCO}_3^-$ , 8 mM | 0.075                          | 1720                                                              | 47.8                              | 0.373             | 149                               |
| <i>Synechococcus</i> 7002, <i>Shewanella</i> W3-18-1 light-limited chemostat  | $\text{HCO}_3^-$ , 8 mM | 0.075                          | 640                                                               | 48.2                              | 0.289             | 116                               |

Abbreviation: AFDW, ash-free dry weight.

<sup>a</sup>The cells were determined to be light saturated and carbon limited when steady-state biomass concentration did not respond to any changes in incident irradiance. Conversely, under light-limited conditions, the biomass concentration changed as a function of irradiance.

<sup>b</sup>The irradiance values for each steady state are reported as quanta incident to the center of the reactor.



disrupt large aggregates and then fixed with 2% paraformaldehyde. Using the 488-nm excitation from a Sapphire LP laser (Coherent Inc., Santa Clara, CA, USA) at 100 mW, samples were analyzed using a 70- $\mu$ m nozzle. Optimization and calibration of the fluorescence-activated cell sorting was performed before each analysis using 3- $\mu$ m Ultra Rainbow Fluorescent Particles (Spherotech, Lake Forest, IL, USA). The ratio of the two distinct populations of cells within a mixed microbial community were identified from 50 000 recorded cells using size and complexity gates using FCS Express (Los Angeles, CA, USA) flow cytometry software.

Microscopic images were acquired on a Zeiss LSM 710 Scanning Confocal Laser Microscope (Carl Zeiss MicroImaging GmbH, Jena, Germany) equipped with a W Plan-Apochromatic 63x/1.0 M27 objective. *Shewanella* W3-18-1 cells were visualized by acridine orange (Invitrogen, Grand Island, NY, USA) at 490–577 nm. *Synechococcus* 7002 was visualized by phycocyaninauto fluorescence measured at 640 nm. Images were processed with Volocity (Perkin Elmer, Waltham, MA, USA) and used to obtain the cell size measurements made along the major and minor axis. Cell volume calculations for each organism were carried out using the equation for a prolate ellipsoid,  $V = \frac{4}{3}\pi a^2 b$ , where  $a$  is the diameter of the minor axis, and  $b$  is the diameter of the major axis. The relative biomass content of each organism in the co-culture was calculated using the equation,  $B_1 = \frac{v_1 r_1}{(v_1 r_1 + v_2 r_2)}$ , where  $v_1$  and  $v_2$  are the cell volumes of each organism,  $r_1$  and  $r_2$  are the measured population ratios.

#### RNA isolation and sequencing

Cells for transcriptomic analysis were harvested from duplicate chemostat cultures by centrifugation at 7000 r.p.m. for 5 min at 4 °C, flash frozen in liquid nitrogen and stored at –80 °C. Total RNA extraction was carried out according to previously published protocols (Beliaev *et al.*, 2005). The quality and integrity of the RNA was assessed on an Agilent 2100 Bioanalyzer and only samples with integrity numbers between 8 and 10 were selected for further analysis (Schroeder *et al.*, 2006). Template complementary DNA was prepared using the Applied Biosystems SOLiD Total RNA-Seq Kit (Life Technologies, Carlsbad, CA, USA) according to the manufacturer's protocol. Sequencing was carried out using the SOLiD 5500XL protocol (Life Technologies). The 50-base sequence reads were mapped to the genomes of *Synechococcus* 7002 (GenBank #: NC\_010475) and *Shewanella* W3-18-1 (GenBank #: NC\_008750\_1) using SOLiDTM LifeScope v. 2.5 software. The normalization of the RNA-seq data was carried out using the RPKM (reads per kilobase per million) calculation (Mortazavi *et al.*, 2008). To allow comparisons between conditions, additional normalization relative to the total number of reads across different samples was performed.

#### Functional enrichment analysis

The transcription data were analyzed using pathway genome databases (PGDBs; <http://pathways.mbl.edu/>) developed for *Synechococcus* 7002 (*Synechococcus*PCC7002Cyc) and *Shewanella* W3-18-1 (*Shew*3181Cyc). These databases contain manually curated gene products, regulon predictions available in RegPrecise (Novichkov *et al.*, 2012), and classification of gene products into BioCyc pathways and GO: Biological Process terms (Caspi *et al.*, 2012), which include functions not covered by the BioCyc pathway schema. Previously developed software (Karp *et al.*, 2010) was used to extract gene products, pathways and regulators for sets of differentially expressed genes. Calculations were done to identify enriched pathways, GO: Biological Processes and transcription factors in the data set using the Fisher's exact parent–child method, which was applied to avoid overrepresentation of parent terms (Grossmann *et al.*, 2007).

## Results and Discussion

#### Metabolic coupling leads to co-culture growth on either inorganic or organic carbon

*Synechococcus* 7002 is an oxygenic photoautotroph that cannot use any exogenous organic carbon compounds other than glycerol for growth under photoheterotrophic or dark respiratory conditions (Lambert and Stevens, 1986). It uses nitrate, ammonium or urea as a nitrogen source, and is auxotrophic for vitamin B<sub>12</sub>, which is required for methionine biosynthesis and photosynthetic activity at low temperatures (Batterton and Vanbaalen, 1971; Rippka *et al.*, 1979). Conversely, *Shewanella* W3-18-1 (Nealson and Scott, 2003) is a facultative anaerobe that possesses a versatile heterotrophic metabolism and can utilize a wide array of carbon sources including various carboxylic acids, carbohydrates, amino acids and nucleic acids, a range of di-peptides as nitrogen sources, and DNA as a phosphorus source (Pinchuk *et al.*, 2008; Rodionov *et al.*, 2010; Rodrigues *et al.*, 2011). *Shewanella* W3-18-1 also requires B<sub>12</sub> for glycerol utilization but is able to circumvent the requirement for methionine biosynthesis by employing B<sub>12</sub>-independent methionine synthase.

To serve as a basis for comparison with interacting populations, both organisms were grown axenically in carbon-limited chemostats using 8 mM NaHCO<sub>3</sub><sup>–</sup> (*Synechococcus* 7002) or 5 mM D,L-Na lactate (*Shewanella* W3-18-1) (Table 1). Under these conditions, *Synechococcus* 7002 accumulated 143 mg l<sup>–1</sup> of ash-free dry weight (AFDW) biomass, whereas aerobic chemostat cultures of *Shewanella* W3-18-1 accumulated 131 mg l<sup>–1</sup> AFDW. Co-culture growth was established upon addition of 0.5 l of lactate-grown *Shewanella* W3-18-1 cells to a 5.5-l continuous culture of *Synechococcus* 7002 and a gradual transition from HCO<sub>3</sub><sup>–</sup> to lactate as the sole source

of carbon. The co-culture grown with lactate was dependent on continuous illumination of the reactor vessel as no growth occurred in the dark when using 100% N<sub>2</sub> as the sparging gas. In the light, there was positive selection for both organisms via the production of CO<sub>2</sub> by *Shewanella* W3-18-1 from lactate. At steady state, the lactate-supplemented co-culture accumulated 172.4 mg l<sup>-1</sup> AFDW biomass, 60% of which was *Synechococcus* 7002 and 40% was *Shewanella* W3-18-1 (Table 1; Supplementary Figure S1). Theoretical yield calculations, performed in consideration of both biomass and extracellular metabolite production, confirmed that 5 mM lactate feed was sufficient to support the measured biomass via CO<sub>2</sub> exchange (Supplementary Material Equations). Notably, biomass yield of *Shewanella* W3-18-1 on lactate was twofold lower during co-cultivation than in the axenic state, thus suggesting changes in C flux distributions.

When the C source was shifted from lactate to bicarbonate, the direction of carbon flux was reversed. That is, growth of *Shewanella* W3-18 was driven by the release of photosynthetically fixed organic carbon by *Synechococcus* 7002. At photon flux of 1720 µmol photons m<sup>-2</sup> s<sup>-1</sup>, the bicarbonate-limited co-culture accumulated 149 mg l<sup>-1</sup> AFDW biomass, of which *Synechococcus* 7002 comprised 99% (Table 1; Supplementary Figure S2). An irradiance decrease from 1720 to 640 µmol photons m<sup>-2</sup> s<sup>-1</sup> led to onset of light limitation and drop of biomass yield to 116 mg l<sup>-1</sup> AFDW (Table 1), although the ratio of *Synechococcus* 7002 to *Shewanella* W3-18 population remained unchanged.

#### Identification of putative interactions through metabolite profiling

The total organic carbon concentrations were similar in filtrates of HCO<sub>3</sub><sup>-</sup> grown axenic *Synechococcus* 7002 cultures (3.9 ± 0.2 mg l<sup>-1</sup>) and *Synechococcus* 7002–*Shewanella* W3-18-1 co-cultures (4.7 ± 0.5 mg l<sup>-1</sup>). NMR analysis of dissolved metabolites excreted by axenic *Synechococcus* 7002 cells identified formate, acetate, lactate and alanine as major extracellular metabolites (Table 2). O<sub>2</sub> concentrations positively affected the extracellular accumulation of amino acids with concentrations of alanine, isoleucine, leucine and valine increasing 1.5- to 3-fold. In co-cultures, formate concentrations remained high, while decreased concentrations of acetate, pyruvate, lactate and glycolate suggested their potential utilization by *Shewanella* W3-18-1 (Table 2). These findings are consistent with the genome annotation and phenotype profiling of *Shewanella* W3-18-1, which utilizes formate only under anoxic conditions using the nitrate-inducible formate dehydrogenase (Rodrigues *et al.*, 2011). Nonetheless, measured decreases in steady-state concentrations of excreted organic acids between the axenic *Synechococcus* 7002 and co-cultures

grown on HCO<sub>3</sub><sup>-</sup> can account for only 185 µg l<sup>-1</sup> of total organic carbon, which constitutes <20% of the *Shewanella* W3-18-1 biomass. Thus, cross-feeding between *Synechococcus* 7002 and *Shewanella* W3-18-1 may occur via metabolic intermediates that are not easily detected by NMR, which include partial breakdown products from complex polymers such as exopolysaccharides and other extracellular polymeric substance components (Belenguer *et al.*, 2006).

#### Transcriptome patterning reveals metabolite exchange and downregulation of oxidative stress response during co-cultivation

A total of 473 and 234 transcripts in *Synechococcus* 7002 and *Shewanella* W3-18-1, respectively, displayed ≥2-fold change in relative mRNA abundances as a result of co-cultivation irrespective of the carbon source. In *Synechococcus* 7002, a large fraction of these transcripts was affiliated with the transport (14%), energy metabolism (8%) and cellular processes (7%) role categories (Figure 1a, Supplementary Table S1). Significant decreases in transcript levels of Fe uptake and acquisition genes (for example, synechobactin biosynthesis) indicated higher availability of this important micronutrient for *Synechococcus* 7002 during co-cultivation. At the same time, increased mRNA abundances for genes encoding functions associated with carbon fixation (that is, RuBisCo, carboxysome components, bicarbonate transporter, the H<sup>+</sup>-translocating NADH-quinone dehydrogenase) were likely due to the inorganic carbon limitation in the co-cultures. Other potentially significant responses to co-cultivation in *Synechococcus* 7002 were associated with programmed cell death and detoxification, as manifested by the decreased transcript levels of a substantial number of one- and two-component toxin–antitoxin systems as well as three oxidative stress-related genes (*SYNPCC7002\_A0320*, *SYNPCC7002\_A2422* and *SYNPCC7002\_G0082*).

In *Shewanella* W3-18-1, the largest group of transcripts displaying ≥2-fold change in relative abundance under all co-culture conditions mapped to the transport and amino-acid biosynthesis categories (Figure 1b, Supplementary Table S2). The specific downregulation of methionine and tryptophan biosynthesis pathways in *Shewanella* W3-18-1 supported that these amino acids are likely to be produced and excreted into the medium by *Synechococcus* 7002 (Table 2). Furthermore, the decrease in transcripts encoding key steps of thiamine pyrophosphate formation (*thiDEG*) and thiamine pyrophosphate-dependent branched chain amino-acid biosynthesis (*ilvI*) in *Shewanella* W3-18-1, across all co-culture conditions suggested the presence of other metabolite exchange pathways.

Interestingly, the co-cultivation led to a decrease in transcript abundance within the oxidative stress

**Table 2** NMR analysis of excreted metabolites by photoautotrophically grown axenic and mixed cultures of *Synechococcus* 7002

| Excreted metabolite | Metabolite concentrations under different growth conditions, $\mu\text{M}^a$                                                          |                                                                                                                                        |                                                                                                                                        |                                                                                                                                                                  |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <i>Synechococcus</i> 7002, dilution rate: $0.075\text{ h}^{-1}$ 1720 $\mu\text{mol photons m}^{-2}\text{ s}^{-1}$ 5% DOT <sup>b</sup> | <i>Synechococcus</i> 7002, dilution rate: $0.075\text{ h}^{-1}$ 1720 $\mu\text{mol photons m}^{-2}\text{ s}^{-1}$ 50% DOT <sup>b</sup> | <i>Synechococcus</i> 7002, dilution rate: $0.075\text{ h}^{-1}$ 1720 $\mu\text{mol photons m}^{-2}\text{ s}^{-1}$ 80% DOT <sup>b</sup> | <i>Synechococcus</i> 7002, <i>Shewanella</i> W3-18-1 dilution rate: $0.075\text{ h}^{-1}$ 1720 $\mu\text{mol photons m}^{-2}\text{ s}^{-1}$ 50% DOT <sup>b</sup> |
| Lactate             | ND                                                                                                                                    | 0.9                                                                                                                                    | 0.7                                                                                                                                    | 0.4                                                                                                                                                              |
| Pyruvate            | 0.5                                                                                                                                   | 1.5                                                                                                                                    | 0.7                                                                                                                                    | ND                                                                                                                                                               |
| Acetate             | 2.9                                                                                                                                   | 6.7                                                                                                                                    | 8.4                                                                                                                                    | 2.9                                                                                                                                                              |
| Formate             | 14.0                                                                                                                                  | 191.1                                                                                                                                  | 134.1                                                                                                                                  | 241.7                                                                                                                                                            |
| Glycolate           | ND                                                                                                                                    | 3.1                                                                                                                                    | 1.2                                                                                                                                    | 2.3                                                                                                                                                              |
| Alanine             | 3.5                                                                                                                                   | 8.9                                                                                                                                    | 4.0                                                                                                                                    | 9.2                                                                                                                                                              |
| Isoleucine          | 0.6                                                                                                                                   | 1.0                                                                                                                                    | 1.6                                                                                                                                    | 1.2                                                                                                                                                              |
| Leucine             | 0.8                                                                                                                                   | 0.9                                                                                                                                    | 0.9                                                                                                                                    | 1.3                                                                                                                                                              |
| Methionine          | 0.1                                                                                                                                   | 0.6                                                                                                                                    | 0.3                                                                                                                                    | 0.4                                                                                                                                                              |
| Threonine           | ND                                                                                                                                    | 0.2                                                                                                                                    | 0.1                                                                                                                                    | ND                                                                                                                                                               |
| Valine              | 0.5                                                                                                                                   | 0.9                                                                                                                                    | 0.8                                                                                                                                    | 1.3                                                                                                                                                              |

Abbreviations: DOT, dissolved  $\text{O}_2$  tension; ND, not determined; NMR, nuclear magnetic resonance spectroscopy.

<sup>a</sup>Each measurement represents an average of four replicates. The s.d. for each data point does not exceed 5%.

<sup>b</sup>DOT value represents % of  $\text{O}_2$  saturation.

response subcategory of *Shewanella* W3-18-1 (Supplementary Table S2), which included two putative redoxin domain genes (*ohrA*, *bcp*) and previously uncharacterized gene cluster (*SputW3181\_1296*–*SputW3181\_1306*) that displays a high degree of similarity to the singlet oxygen ( $^1\text{O}_2$ ) protection ChrR regulon of *Rhodobacter sphaeroides* (Dufour *et al.*, 2008). Other significantly downregulated transcripts, whose presence was previously linked to stress response in *Shewanella* (Qiu *et al.*, 2005; McLean *et al.*, 2008), encoded lambda-like bacteriophage proteins and the LrgAB system that controls autolysis via modulating murein hydrolase activity (Groicher *et al.*, 2000). Notably, this coincided with transcriptional upregulation of several surface attachment factors (*pilA*, *pilM*, *pilXWV*, *bpfA* and *csgA*) of *Shewanella* W3-18-1, which are typically associated with biofilm lifestyle (Schembri *et al.*, 2003; Beloin and Ghigo, 2005).

#### Transcriptional response to carbon limitation in a co-culture leads to the maximization of metabolic capacity

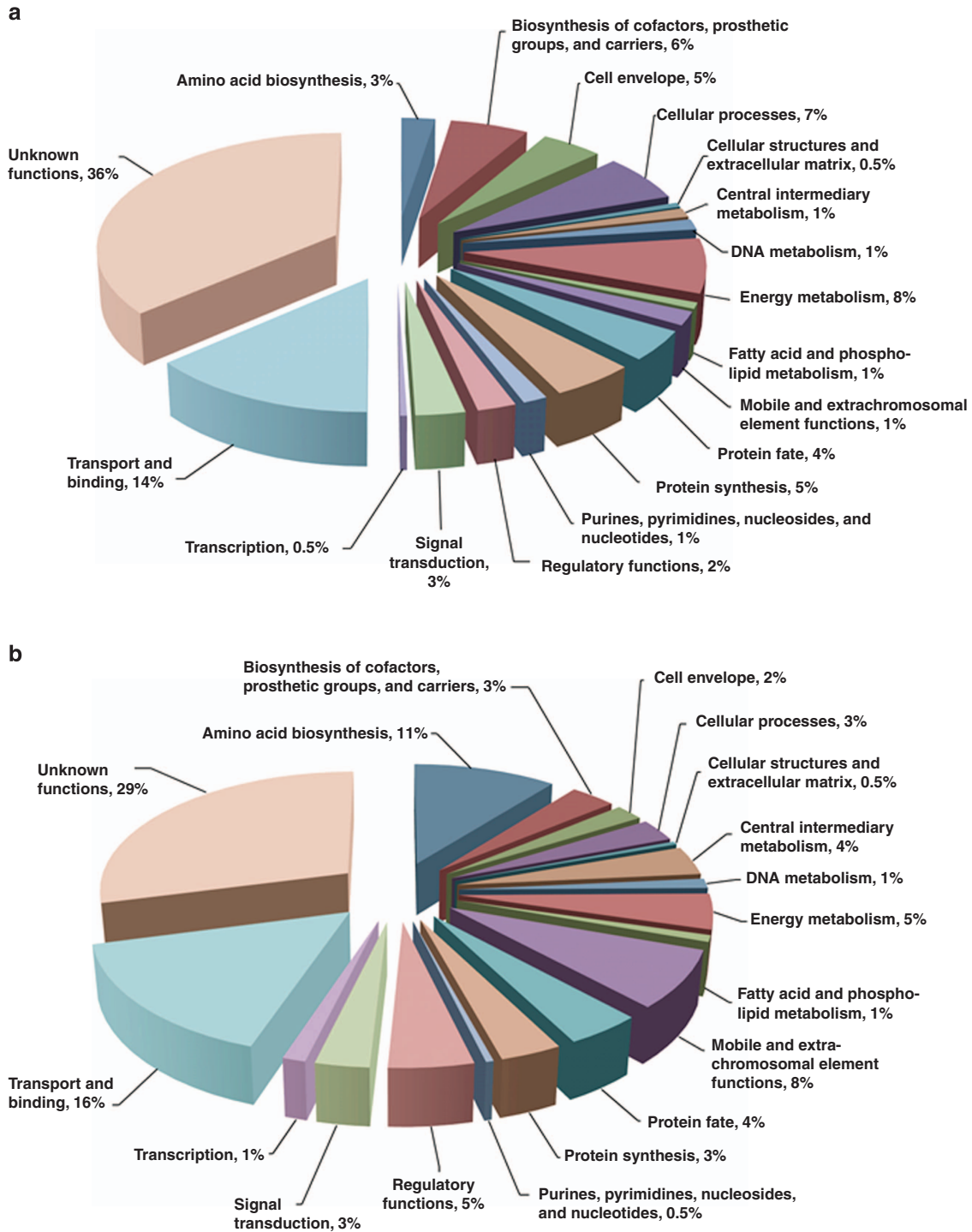
As the growth-limiting carbon source determined flux directions and densities of interacting populations, it exerted a major effect on the global transcriptome patterns in both organisms (Figure 2). During lactate-supported growth, 1184 *Synechococcus* 7002 genes and 456 *Shewanella* W3-18-1 genes showed  $\geq 2$ -fold change in relative mRNA abundances (Supplementary Tables S3 and S4). Conversely, in  $\text{HCO}_3^-$  grown co-cultures, 690 *Synechococcus* 7002 genes and 1342 *Shewanella* W3-18-1 genes were differentially expressed (Supplementary Tables S5 and S6). The inverted patterns likely reflect substrate availability and flux partitioning as carbon limitation, and to a larger degree carbon starvation, lead to competitive

inter-specific interactions in mixed cultures (Kjelleberg *et al.*, 1993; Garbeva and de Boer, 2009).

Pathway enrichment analysis comparing global transcript levels in lactate-grown co-cultures relative to axenic controls indicated that both *Synechococcus* 7002 and *Shewanella* W3-18-1 had maximized the uptake and utilization of specific nutrients during co-cultivation. In *Synechococcus* 7002 (Table 3, Supplementary Table S3), the increase in mRNA levels of putative glyoxylate metabolism enzymes, for example, D-3-phosphoglycerate dehydrogenase (*SYNPCC7002\_A1246*) and phosphoglycolate phosphatase (*SYNPCC7002\_A0506*), suggested occurrence of photorespiration, which was likely due to the elevated  $\text{O}_2/\text{CO}_2$  ratio compared with bicarbonate-grown cultures (Table 1). Interestingly, despite the obvious excess of ammonium in the growth medium, a broad transcriptional upregulation of putative nitrogen assimilation machinery in *Synechococcus* 7002 was observed, which included glutamine synthetase (*glnA*) and assimilatory ferredoxin-nitrite reductase (*nirA*) genes. In addition, increased transcript abundance of the L-arginine deiminase (*arcA*), and L-asparagine amidopeptidase (*asnA1*) genes, which are involved in cyanophycin utilization, also suggests utilization of nitrogen-containing organic compounds as an additional source of carbon by the cyanobacterium.

In *Shewanella* W3-18-1 (Table 4; Supplementary Table S4), pathway enrichment analysis revealed that growth in lactate-supplemented co-cultures was also associated with elevated transcription of nitrogen transport and assimilation genes. Specifically, the relative mRNA abundances of nitrogen regulator (*glnK*), glutamine synthetase (*glnA*) and ammonia transporter (*amt*) genes increased 25-, 5- and 8-fold, respectively, relative to the axenic state. Broad decrease in transcript abundance was also displayed by pathways involved in biosynthesis of amino acid,





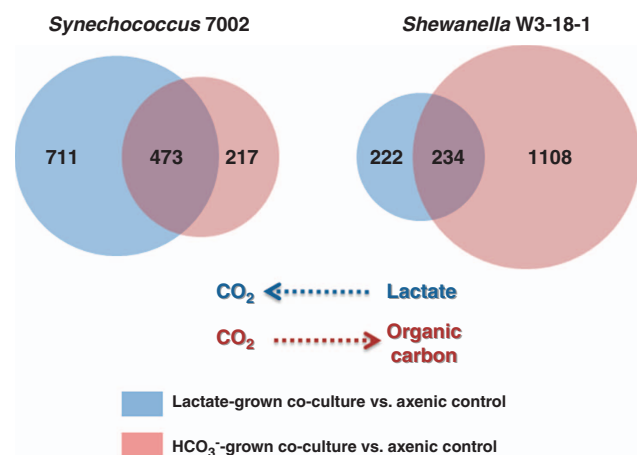
**Figure 1** Functional assignments of (a) *Synechococcus* 7002 and (b) *Shewanella* W3-18-1 genes displaying  $\geq 2$ -fold change in mRNA abundances under all co-culture conditions relative to the axenic state. The transcriptome comparisons were conducted using co-cultures and axenic cells grown in carbon-limited chemostats at a photon flux of  $1720 \mu\text{mol m}^{-2} \text{s}^{-1}$ .

such as methionine, tryptophan, isoleucine and valine. At the same time in lactate-supplemented co-cultures, *Shewanella* W3-18-1 downregulated transcripts for genes encoding D,L-lactate (*lldP*, *lldEF* and *SputW3181\_2834*) utilization machinery (Pinchuk *et al.*, 2009), suggesting preferential utilization of organic compounds excreted or released by

the cyanobacterium. Supporting this notion were elevated mRNA levels of alanine dehydrogenase (Ald) and putative nucleotide deaminase (W3181\_3499), which can provide *Shewanella* W3-18-1 with additional sources of carbon and nitrogen (Supplementary Table S4). Upregulation of extracellular phospholipase A (*pldA*), lipase (W3181\_1613) and malate synthase

(*aceB*) genes can potentially increase carbon availability via the degradation of extracellular lipopolysaccharides into acetyl-CoA and subsequent incorporation into the tricarboxylic acid cycle.

The opportunistic nature of interactions was even more evident during co-culturing using  $\text{HCO}_3^-$  as the sole source of C. Under these conditions, the only influx of organic carbon for heterotrophic growth of *Shewanella* W3-18-1 was provided by the cyanobacterium, either by excretion, cell lysis or exoenzyme-mediated release from the cyanobacterial cell surfaces. When compared with lactate-grown co-cultures (Supplementary Table S7), *Synechococcus*



**Figure 2** Venn diagram of the global transcriptome changes in *Synechococcus* 7002 and *Shewanella* W3-18-1 in response to co-cultivation. Arrows indicate the direction of the carbon flux in lactate (blue) and  $\text{HCO}_3^-$  (red)-supplemented co-cultures. The area of each circle represents the total number of genes displaying  $\geq 2$ -fold change in mRNA abundances in co-cultures relative to the axenic state. The transcriptome comparisons were conducted using co-cultures and axenic cells grown in carbon-limited chemostats at a photon flux of  $1720 \mu\text{mol m}^{-2} \text{s}^{-1}$ .

7002 transcription profiles displayed down-regulation of photorespiration and carbohydrate metabolism genes during co-culturing with  $\text{HCO}_3^-$ . These patterns as well as elevated transcript levels of Fe acquisition genes (Table 5) are more likely to be linked to the increased production of light-capturing and photosynthetic machinery (Ludwig and Bryant, 2012; Morrissey and Bowler, 2012) and higher  $\text{CO}_2$  availability (Table 1). Similarly, lower mRNA levels of putative nitrile/cyanate, dipeptide and amino acids transporter genes suggested decreased need for additional (organic) carbon sources during photoautotrophic growth of *Synechococcus* 7002.

Increased transcription levels of multiple catabolic pathways and the putative carbon starvation protein (*cstA*) involved in peptide uptake was indicative that *Shewanella* W3-18-1 experienced severe C-limitation upon shift of the co-culture from lactate to  $\text{HCO}_3^-$  (Table 6; Supplementary Table S8). Elevated transcript levels were detected for genes involved in lipid, amino acid (Phe, Tyr and Met), polyamine, putrescine and organic acids (acetate, acetoacetate and propionate) utilization. Furthermore, enrichment analysis of predicted regulons verified that metabolism of fatty acids (*fadR*, *fabR* and *psrA*), and degradation of histidine, tyrosine and propionate (*hutC*, *tyrR* and *prpR*) were significantly upregulated in *Shewanella* W3-18-1 during growth in  $\text{HCO}_3^-$ -supplemented co-cultures (Table 6).

#### Cell envelope modifications and evidence of cell-cell interactions

During co-culture growth on lactate, where *Shewanella* W3-18-1 accounted for  $> 80\%$  of the co-culture cell population (Supplementary Figure S1), microscopic observations revealed the presence of multicellular aggregates comprised of cyanobacterial and proteobacterial cells (Figure 3). It is important to

**Table 3** Pathway and regulator enrichment for *Synechococcus* 7002 grown in carbon-limited chemostat co-culture with lactate as the sole source of carbon

| Pathway/regulon enrichment                                                                                                                                                                                                                                                                                                                                                                                                                                        | P-value <sup>a</sup> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <i>Genes displaying <math>\geq 2</math>-fold increase in mRNA abundance relative to the axenic state</i>                                                                                                                                                                                                                                                                                                                                                          |                      |
| Cell structures biosynthesis: <i>ddl</i> , <i>glmM</i> , <i>lpxA</i> , <i>mraY</i> , <i>murA</i> , <i>murC</i> , <i>murE</i> , <i>murF</i> , <i>murG</i> , <i>murI</i> , <i>SYNPCC7002_A0138</i>                                                                                                                                                                                                                                                                  | 0.0075               |
| Fatty acid biosynthesis, initiation and elongation: <i>fabB</i> , <i>fabF</i> , <i>fabZ</i>                                                                                                                                                                                                                                                                                                                                                                       | 0.0220               |
| UDP-N-acetylmuramoyl-pentapeptide biosynthesis: <i>ddl</i> , <i>murA</i> , <i>murC</i> , <i>murE</i> , <i>murF</i> , <i>murI</i>                                                                                                                                                                                                                                                                                                                                  | 0.0221               |
| Photorespiration: <i>gcvT</i> , <i>glcE</i> , <i>rbcs</i> , <i>serA</i> , <i>SYNPCC7002_A0506</i>                                                                                                                                                                                                                                                                                                                                                                 | 0.0228               |
| Vitamin biosynthesis: <i>bioD</i> , <i>bioF</i> , <i>coaE</i> , <i>cobP</i> , <i>cobT</i> , <i>fabB</i> , <i>fabF</i> , <i>fabZ</i> , <i>folE</i> , <i>folQ</i> , <i>gcvT</i> , <i>hggT</i> , <i>ilvC</i> , <i>panB</i> , <i>pdxH</i> , <i>plr1</i> , <i>purU</i> , <i>pyrR</i> , <i>ribAB</i> , <i>serA</i> , <i>SYNPCC7002_A0363</i> , <i>vte1</i> , <i>ygfA</i>                                                                                                | 0.0240               |
| Nucleic acid processing: <i>queA</i> , <i>queF</i> , <i>tgt</i>                                                                                                                                                                                                                                                                                                                                                                                                   | 0.0277               |
| <i>Genes displaying <math>\geq 2</math>-fold decrease in mRNA abundance relative to the axenic state</i>                                                                                                                                                                                                                                                                                                                                                          |                      |
| Siderophore biosynthesis: <i>sybA</i> , <i>sybB</i> , <i>sybC</i> , <i>sybD</i> , <i>sybE</i> , <i>sybF</i> , <i>sybL</i>                                                                                                                                                                                                                                                                                                                                         | 0.0000               |
| Oxygenic photosynthesis: <i>petG</i> , <i>petH</i> , <i>petM</i> , <i>prk</i> , <i>psaE</i> , <i>psaK</i> , <i>psbD</i> , <i>psbE</i> , <i>psbF</i> , <i>psbJ</i> , <i>psbL</i> , <i>psbO</i> , <i>psbT</i> , <i>psbY</i> , <i>rpiA</i>                                                                                                                                                                                                                           | 0.0062               |
| Photosynthesis, light reactions: <i>petG</i> , <i>petH</i> , <i>petM</i> , <i>psaE</i> , <i>psaK</i> , <i>psbD</i> , <i>psbE</i> , <i>psbF</i> , <i>psbJ</i> , <i>psbL</i> , <i>psbO</i> , <i>psbT</i> , <i>psbY</i>                                                                                                                                                                                                                                              | 0.0083               |
| Generation of precursor metabolites and energy: <i>ald</i> , <i>ctaCl</i> , <i>fumC</i> , <i>gap</i> , <i>hoxH</i> , <i>hoxY</i> , <i>ndhD1</i> , <i>ndhM</i> , <i>nifj</i> , <i>petG</i> , <i>petH</i> , <i>petM</i> , <i>ppsA</i> , <i>prk</i> , <i>psaE</i> , <i>psaK</i> , <i>psbD</i> , <i>psbE</i> , <i>psbF</i> , <i>psbJ</i> , <i>psbL</i> , <i>psbO</i> , <i>psbT</i> , <i>psbY</i> , <i>rpiA</i> , <i>sdhB</i> , <i>talA</i> , <i>talC</i> , <i>zwf</i> | 0.0240               |
| Ferric uptake, Fur regulon: <i>chlL</i> , <i>chlN</i> , <i>exbB</i> , <i>exbD</i> , <i>fecB</i> , <i>fecC</i> , <i>fecD</i> , <i>fecE</i> , <i>fhuA</i> , <i>futC</i> , <i>hik20</i> , <i>nifj</i> , <i>pchR</i> , <i>scht</i> , <i>SYNPCC7002_A2346</i> , <i>SYNPCC7002_G0005</i> , <i>SYNPCC7002_G0006</i> , <i>SYNPCC7002_G0089</i> , <i>SYNPCC7002_G0099</i> , <i>tonB</i>                                                                                    | 0.0006               |

<sup>a</sup>P-value represents the probability that the number of genes associated with a specific pathway/regulon occurs by chance.



**Table 4** Pathway and regulator enrichment for *Shewanella* W3-18-1 in carbon-limited chemostat co-culture with lactate as the sole source of carbon

| Pathway/regulon enrichment                                                                                                                                                           | P-value <sup>a</sup> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <i>Genes displaying ≥2-fold increase in mRNA abundance relative to the axenic state</i>                                                                                              |                      |
| Amino acids biosynthesis: <i>argC, aroF, asnB, cysD, cysJ, glnA, hisD, hisG, lysC, tyrA</i>                                                                                          | 0.0001               |
| Metabolism of inorganic nutrients: <i>cysD, cysJ, glnA, nrfA</i>                                                                                                                     | 0.0063               |
| Phospholipase: <i>pldA</i>                                                                                                                                                           | 0.0130               |
| Alanine degradation: <i>ald</i>                                                                                                                                                      | 0.0133               |
| Nitrogen assimilation, NtrC regulon: <i>amtB, glnA, glnK, SputW3181_0223</i>                                                                                                         | 0.0001               |
| Tryptophan biosynthesis, TrpR regulon: <i>aroF, tyrA</i>                                                                                                                             | 0.0043               |
| <i>Genes displaying ≥2-fold decrease in mRNA abundance relative to the axenic state</i>                                                                                              |                      |
| Aspartate super-pathway: <i>asd, lysA, metA, metB, metC, metE, metH, metK, metL</i>                                                                                                  | 0.0001               |
| Threonine degradation: <i>ilvA, ilvC, ilvD, ilvG, ilvH, ilvI</i>                                                                                                                     | 0.0002               |
| Tryptophan biosynthesis: <i>trpA, trpB, trpCF, trpD, trpE, trpG</i>                                                                                                                  | 0.0002               |
| Fatty acid and lipids degradation: <i>fadA, fadB, fadE, fadI, fadJ, glpD, SputW3181_1978, SputW3181_2433</i>                                                                         | 0.0004               |
| Homoserine and methionine biosynthesis: <i>asd, metA, metB, metC, metE, metH, metL</i>                                                                                               | 0.0069               |
| Isoleucine and valine biosynthesis: <i>asd, ilvA, ilvC, ilvD, ilvG, ilvH, ilvI, metL</i>                                                                                             | 0.0106               |
| B <sub>12</sub> biosynthesis and uptake: <i>cobC, cobS, cobT, cobU, btuC, btuD</i>                                                                                                   | 0.0237               |
| Thiamin biosynthesis <i>thiC, thiDE, thiF, thiH</i>                                                                                                                                  | 0.0237               |
| Superpathway of chorismate metabolism: <i>aroE, pheA, trpA, trpB, trpCF, trpD, trpE, trpG, ubiC</i>                                                                                  | 0.0276               |
| Methylglyoxal detoxification: <i>dldD, gloB</i>                                                                                                                                      | 0.0286               |
| Methionine biosynthesis, MetJ regulon: <i>btuB, metA, metB, metE, metF, metH, metK, metL, metR, metT, metY, msaA, SputW3181_0088, SputW3181_0089, SputW3181_1636, SputW3181_2707</i> | 0.0000               |
| Fatty acid degradation, PsrA regulon: <i>etfQ, fadA, fadB, fadE, fadH, fadI, fadJ, psrA, SputW3181_1978, SputW3181_2146</i>                                                          | 0.0004               |
| Methionine biosynthesis, MetR regulon: <i>metC, metE, metR</i>                                                                                                                       | 0.0058               |
| Fatty acid degradation, FadR regulon: <i>fadI, fadJ, fadL</i>                                                                                                                        | 0.01319              |
| Lactate utilization, LldR regulon: <i>lldE, lldF</i>                                                                                                                                 | 0.03802              |

<sup>a</sup>P-value represents the probability that the number of genes associated with a specific pathway/regulon occurs by chance.

**Table 5** Pathway and regulator enrichment for *Synechococcus* 7002 grown in carbon-limited chemostat co-culture with HCO<sub>3</sub><sup>−</sup> as the sole source of carbon

| Pathway/regulon enrichment                                                                                  | P-value <sup>a</sup> |
|-------------------------------------------------------------------------------------------------------------|----------------------|
| <i>Genes displaying ≥2-fold increase in mRNA abundance relative to the lactate co-culture</i>               |                      |
| Histidine biosynthesis: <i>hisA, hisB, hisF</i>                                                             | 0.0431               |
| Ferric uptake, Fur regulon: <i>exbB, fecB, fhuA, hik20, pchR, SYNPC7002_G0005, SYNPC7002_G0006</i>          | 0.0718               |
| Heat shock response, HrcA regulon: <i>groES, SYNPC7002_A0107</i>                                            | 0.0220               |
| <i>Genes displaying ≥2-fold decrease in mRNA abundance relative to the lactate co-culture</i>               |                      |
| Photorespiration: <i>gcvT, rbcS, serA, SYNPC7002_A0506</i>                                                  | 0.0033               |
| Carbohydrates metabolism: <i>glgA2, gpmB, rbcS, sppA, SYNPC7002_A0448, SYNPC7002_A1492, SYNPC7002_A1554</i> | 0.0407               |
| Serine biosynthesis: <i>serA</i>                                                                            | 0.0045               |

<sup>a</sup>P-value represents the probability that the number of genes associated with a specific pathway/regulon occurs by chance.

note that, while heterogeneity across the aggregates may exert some effects on the co-culture transcriptome patterns, these effects are likely to be minimal. For *Synechococcus* 7002, only a substantially smaller portion of the cells were attached to the aggregates of *Shewanella* W3-18-1, and then were located at the surface (Figure 3). For *Shewanella* W3-18-1, which also aggregated when grown axenically under identical conditions (data not shown), cells in the interior of the flocs may be growing more slowly than those at the periphery or those suspended singly; however, to the extent that this is true, chemostat theory predicts a compensation in which other cells are growing faster than the dilution rate.

In lactate-grown co-cultures, enrichment of pathways involved in biosynthesis of fatty acids, peptidoglycan and lipopolysaccharides indicated

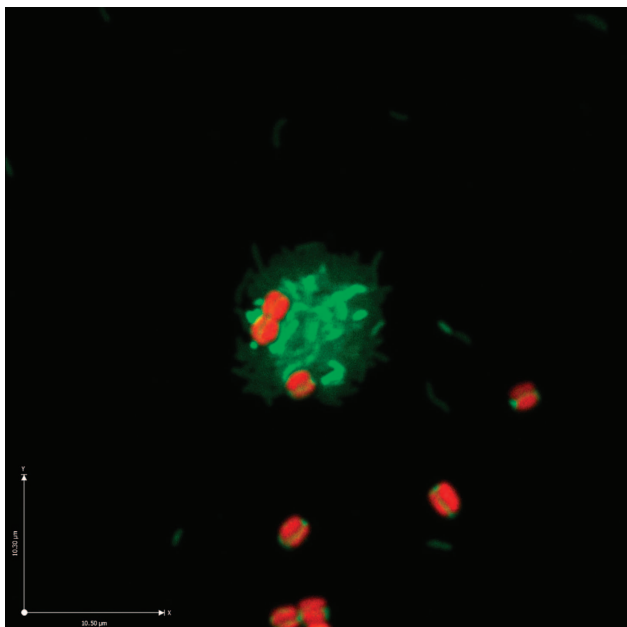
remodeling of the cell surface in *Synechococcus* 7002. Transcripts encoding the *N*-acetyl muramoyl-pentapeptide biosynthesis reactions (*ddl, murA, murC, murE, murF* and *murI*; Table 3; Supplementary Table S3) were upregulated, while those encoding putative transcarboxypeptidases (*dacB, dacC, dacD* and *SynPCC7002\_A0082*) and lipopolysaccharide biosynthesis enzymes were broadly downregulated in *Synechococcus* 7002. In *Shewanella* W3-18-1, a number of biofilm formation factors including type IV pili (*pilAB, pilWWV* and *pilM*), curli (*csgAB, csgD* and *csgEFG*) and extracellular proteins predicted to promote adhesion (*pilY, bpfA, Sputw3181\_3687* and *SputW3181\_1756*) also displayed elevated mRNA levels under co-culture conditions (Supplementary Table S4).

Furthermore, the observed cell–cell interactions between the two organisms could facilitate the

**Table 6** Pathway and regulator enrichment for *Shewanella* W3-18-1 in carbon-limited chemostat co-culture with  $\text{HCO}_3^-$  as the sole source of carbon

| Pathway/regulon enrichment                                                                                                                                                                                                                                                                                                 | P-value <sup>a</sup> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <i>Genes displaying <math>\geq 2</math>-fold increase in mRNA abundance relative to the lactate co-culture</i>                                                                                                                                                                                                             |                      |
| Amino acids biosynthesis: <i>aroA, aroE, carA, hisA, hisB, hisC, hisD, hisF, hisG, hisH, hisIE, ilvA, ilvD, ilvE, ilvG, ilvM, kbl, metE, phhA, serB, thrA, thrB, thrC, trpB, trpG</i>                                                                                                                                      | 0.0000               |
| Degradation/utilization/assimilation: <i>acnD, acs, aguA, can, deoD, fadA, fadB, fadD, fadE, fadI, fadJ, gabD, gloB, hmgA, hppD, hutH, hutU, ilvA, ilvD, ilvE, ilvG, ilvM, kbl, liuF, liuG, mdeA, phhA, phhB, prpB, prpC, prpF, puuA, puuB, puuC, speC, SputW3181_1978, SputW3181_2433, SputW3181_4006, SputW3181_4017</i> | 0.0001               |
| Fatty acid and lipids degradation: <i>fadA, fadB, fadD, fadE, fadI, fadJ, liuF, liuG, SputW3181_1978, SputW3181_2433, SputW3181_4017</i>                                                                                                                                                                                   | 0.0001               |
| Glyoxylate cycle: <i>aceA, aceB, gltA, sdhC, sdhD</i>                                                                                                                                                                                                                                                                      | 0.0003               |
| Guanosine nucleotides <i>de novo</i> biosynthesis: <i>gmk, guaB, ndk, nrdA</i>                                                                                                                                                                                                                                             | 0.0207               |
| Fatty acid degradation, PsrA regulon: <i>aceA, aceB, etfQ, fadA, fadB, fadD, fadE, fadH, fadI, fadJ, gltA, psrA, sdhC, sdhD, SputW3181_1978, SputW3181_2146</i>                                                                                                                                                            | 0.0001               |
| Fatty acid biosynthesis repressor, FabR regulon: <i>desC, fabR, fadL, plsC, SputW3181_4017, yqfA</i>                                                                                                                                                                                                                       | 0.0001               |
| Methyl citrate utilization repressor, PrpR regulon: <i>acnD, prpB, prpC, prpF</i>                                                                                                                                                                                                                                          | 0.0004               |
| Histidine utilization repressor, HutC regulon: <i>hutC, hutH, hutU</i>                                                                                                                                                                                                                                                     | 0.0104               |
| Amino acid (Tyr/Phe) utilization activator, TyrR regulon: <i>aceA, aceB, aroA, can, liuF, liuG, mdeA, phhA, phhB, SputW3181_1883</i>                                                                                                                                                                                       | 0.0153               |
| Fatty acid degradation, FadR regulon: <i>fadI, fadJ, fadL</i>                                                                                                                                                                                                                                                              | 0.0233               |
| <i>Genes displaying <math>\geq 2</math>-fold decrease in mRNA abundance relative to the lactate co-culture</i>                                                                                                                                                                                                             |                      |
| Thiamin biosynthesis: <i>sufS, thiC, thiDE, thiF, thiH, thiL</i>                                                                                                                                                                                                                                                           | 0.0001               |
| Inorganic nutrients metabolism: <i>cysC, cysD, cysH, cysI, cysJ, cysN, nrfA, phsB</i>                                                                                                                                                                                                                                      | 0.0029               |
| Secondary metabolites degradation: <i>aceE, aceF, lldE, lldF, lldG, nagA</i>                                                                                                                                                                                                                                               | 0.0077               |
| Pentose phosphate pathway, oxidative branch: <i>gnd, pgl, zwf</i>                                                                                                                                                                                                                                                          | 0.0286               |
| Methionine biosynthesis: <i>cysC, cysD, cysH, cysI, cysJ, cysN, metE, metK, metL</i>                                                                                                                                                                                                                                       | 0.0287               |
| Arginine biosynthesis: <i>argB, argC, argF, argG, argH</i>                                                                                                                                                                                                                                                                 | 0.0332               |
| Rhamnose biosynthesis: <i>rbfC, rmlC, rmlD</i>                                                                                                                                                                                                                                                                             | 0.0350               |
| Lactate utilization, LldR regulon: <i>lldE, lldF, lldG</i>                                                                                                                                                                                                                                                                 | 0.0057               |
| Pyruvate metabolism repressor, PdhR regulon: <i>aceE, aceF, oadA, oadB, pflA, flB, pflX</i>                                                                                                                                                                                                                                | 0.0018               |
| Tryptophan biosynthesis, TrpR regulon: <i>aroF, trpR, tyrA</i>                                                                                                                                                                                                                                                             | 0.0057               |

<sup>a</sup>P-value represents the probability that the number of genes associated with a specific pathway/regulon occurs by chance.



**Figure 3** Representative micrograph of *Synechococcus* 7002 and *Shewanella* W3-18-1 cell aggregates formed in a co-culture grown under carbon-limited aerobic chemostat conditions using 5 mM lactate as the sole source of carbon. *Synechococcus* 7002 cells (red) were visualized by phycocyanin autofluorescence measured at 640 nm. *Shewanella* W3-18-1 cells (green) were visualized by acridine orange staining and detection at 490–577 nm.

uptake and re-utilization of complex exopolymers or organic substrates released from the surface. In  $\text{HCO}_3^-$  grown co-cultures (Supplementary Table S4), *Shewanella* W3-18-1 displayed increased transcript abundances of genes encoding secreted or outer membrane proteins including a lipase (*SputW3181\_1613*), several proteases (*sapSH*, *Sputw3181\_3341*, *Sputw3181\_3384* and *SputW3181\_0531*) and a polymorphic toxin (*Sputw3181\_0994*), which together may be involved in cell envelope lysis and degradation of surface structures (Zhang *et al.*, 2012). Similarly, the elevated mRNA levels of 4- $\alpha$ -glucanotransferase gene (*malQ*) can potentially allow for carbohydrate utilization, while DNA can be broken down by the putative extracellular endonuclease (*exeS*). Proteins/peptides appeared also to be used by *Shewanella* W3-18-1 as well, perhaps directly as amino acids or after further catabolism as a nitrogen, carbon and energy source (Supplementary Table S4).

## Conclusions

Development of predictive, system-level understanding of microbial consortia requires integration of authentic field investigation with well-controlled laboratory-scale studies. Here, we applied deep

transcriptome sequencing to a laboratory-based model system consisting of a euryhaline cyanobacterium, *Synechococcus* 7002 and a marine heterotroph, *Shewanella* W3-18-1, to investigate the effect of co-cultivation and carbon flux directions on the interactions between these organisms. Although the opportunistic nature of this association constrains, to some degree, the ecological implications of observed transcriptional responses, the results provide a number of novel and relevant insights into the physiological basis of microbial interactions.

The overall transcriptional response of both organisms to co-cultivation was shaped by their respective physiologies and the growth constraints. Carbon limitation resulted in the expansion of metabolic capacity, which was manifested through transcriptional upregulation of transport and catabolic pathways (Tables 3–6). As metabolic coupling occurred either via lactate oxidation or secretion of photosynthetically fixed carbon, NMR (Table 2) and transcriptome analyses (Tables 3–6) suggested specific metabolite exchange during co-cultivation. On one hand, the detection of methionine and alanine excretion by the cyanobacterium leads us to hypothesize that downregulation of the corresponding biosynthetic machinery of *Shewanella* W3-18-1 reflects utilization of these amino acids by the heterotroph. On the other hand, the broad and consistent decrease of mRNA levels for many Fe-regulated *Synechococcus* 7002 genes during co-cultivation, including synechobactin biosynthesis pathway and the Fur regulon (Table 2 and Supplementary Table S4), suggested increased Fe availability in the cyanobacterium during co-cultivation.

The latter observation is in agreement with a previous study, in which downregulation of Fe-acquisition machinery was reported in *Synechococcus* sp. WH8102 during co-cultivation with *Vibrio parahaemolyticus* (Tai *et al.*, 2009). However, the causative effect of co-cultivation on Fe homeostasis in cyanobacteria is yet to be determined. Although increased Fe availability can be a result of lower demand—for instance, owing to declining photosynthetic activity (Ludwig and Bryant, 2012), the mRNA abundance of *Synechococcus* 7002 genes involved in photosynthetic metabolism was high across all experimental conditions. A more compelling explanation may be linked to the documented capacity of aerobic heterotrophic microorganisms to produce large quantities of extracellular Fe chelators (Wandersman and Delepelaire, 2004), which can be recognized and transported intracellularly by the cyanobacterium. Bacteria of the genus *Shewanella* were also shown to carry out Fe reduction aerobically inside cell aggregates (McLean *et al.*, 2008) thus providing more facile and energy-efficient mechanisms for Fe acquisition by the cyanobacterium.

Furthermore, this study provides evidence of potentially novel interactions between oxygenic photoautotrophs and heterotrophs related to the oxidative stress response. Our results are distinct

from those reported in a previous *Prochlorococcus* spp. study, where heterotrophs were hypothesized to decrease oxidative stress in the cyanobacterium via catalase-dependent ROS scavenging (Morris *et al.*, 2008). In *Shewanella* W3-18-1, transcript levels of genes involved in oxidative stress response and scavenging of ROS radicals displayed broad decrease under co-culture conditions, which included putative  $^1\text{O}_2$  protection (*chrR* regulon) (Dufour *et al.*, 2008) and methylglyoxal degradation (Hoque *et al.*, 2010) pathways (Table 4 and Supplementary Table S4). Because of the propensity of marine heterotrophs to produce large quantities of extracellular superoxide and related ROS (Diaz *et al.*, 2013), our observations indicate that some photoautotrophs may provide community-level protection against oxidative stress as opposed to being protected by other members. In that regard, our study not only measures the responses at the system-level to explore the interactions between the partners more holistically but also formulates new hypotheses about cyanobacterial–heterotrophic interactions.

## Conflict of Interest

The authors declare no conflict of interest.

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## Database access to the sequencing data

The RNA sequencing data are available online at the Gene Expression Omnibus (GEO) database (<http://www.ncbi.nlm.nih.gov/gds/>) under accession number GSE53360.

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